# PQ-Auth Sentinel — Research Testing Report

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## Methodology

The CSV is split into 60% training, 20% validation and 20% held-out testing using stratification. Candidate classifiers are compared on validation macro-F1; the selected model is then reported on the held-out test set. Leakage-prone output fields are excluded from the feature matrix.

## Selected model and held-out results

| Measure | Result |
|---|---|
| Selected algorithm | Soft Voting (Extra Trees + Random Forest) |
| Test accuracy | 99.05% |
| Test macro-F1 | 0.990 |
| Test rows | 4000 |
| Dataset rows used | 20000 |

## Algorithm comparison

| Algorithm | Val accuracy | Val macro-F1 | Test accuracy | Test macro-F1 |
|---|---|---|---|---|
| Logistic Regression | 0.992 | 0.991 | 0.987 | 0.987 |
| Decision Tree | 0.986 | 0.986 | 0.987 | 0.987 |
| Random Forest | 0.992 | 0.992 | 0.990 | 0.989 |
| Extra Trees | 0.992 | 0.992 | 0.991 | 0.991 |
| Hist Gradient Boosting | 0.992 | 0.992 | 0.991 | 0.990 |
| Soft Voting (Extra Trees + Random Forest) | 0.993 | 0.992 | 0.991 | 0.990 |

## Scenario results

| ID | Scenario | Expected class | ML prediction | Risk proxy | Policy suggestion |
|---|---|---|---|---|---|
| SC-01 | Normal authentication | Normal | Normal | 0.000 | ALLOW |
| SC-02 | Replay-like activity | Replay | Spoofing | 1.000 | BLOCK |
| SC-03 | Flooding-like activity | Flooding | Spoofing | 1.000 | BLOCK |
| SC-04 | Spoofing-like behaviour | Spoofing | Spoofing | 1.000 | BLOCK |
| SC-05 | Low-rate suspicious activity | Suspicious | Spoofing | 1.000 | BLOCK |
| SC-06 | Repeated failed attempts | Suspicious | Spoofing | 1.000 | BLOCK |
| SC-07 | Untrusted certificate context | Spoofing | Spoofing | 0.994 | BLOCK |

## Cryptographic smoke tests

| Test | Result | Detail |
|---|---|---|
| ML-DSA-65 availability | PASS | Algorithm enabled. |
| Valid signature verification | PASS | Sign then verify canonical test bytes. |
| Tampered message rejected | PASS | Modified message should not verify. |

## Approval recommendation

For academic research review only. Model metrics are measured on the supplied dataset and may reflect synthetic data patterns. Scenario profiles are controlled inputs. The report does not certify production security. Before deployment, perform independent review, calibrate risk thresholds, use durable replay state, and validate against representative real-world traffic.